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A. Mutual k-Nearest Neighbor Alignment Metric

For two models with representations f, g the mutual k-nearest neighbor metric measures the average overlap of their respective nearest neighbor sets. In this section, we refer to this metric as m_{NN}, which we will formally define below.

For cross-modal domains, define $(x_i, y_i) \in \mathcal{X}$ as a sample from the data distribution $\mathcal{X}$ (*e.g.* image-caption dataset). For the single domain alignment measurements, the samples are equivalent $x_i = y_i$ (*e.g.*, images for vision, and text for language). Let $\{x_i, y_i\}_{i=1}^b$ be the corresponding mini-batch sampled from this data distribution. Then given two model representations f and g the corresponding features are: $\phi_i = f(x_i)$ and $\psi_i = g(y_i)$, where the collection of these features are denoted as $\Phi = \{\phi_1, \ldots, \phi_b\}$ and $\Psi = \{\psi_1, \ldots, \psi_b\}$. Then for each feature pair (ϕ_i, ψ_i), we compute the respective nearest neighbor sets $\mathcal{S}(\phi_i)$ and $\mathcal{S}(\psi_i)$.

$$d_{\mathsf{knn}}(\phi_i, \Phi \setminus \phi_i) = \mathcal{S}(\phi_i) \tag{9}$$

$$d_{\mathsf{knn}}(\psi_i, \Psi \setminus \psi_i) = \mathcal{S}(\psi_i) \tag{10}$$

where d_{knn} returns the set of indices of its k-nearest neighbors. Then we measure its average intersection via

$$m_{\mathsf{NN}}(\phi_i, \psi_i) = \frac{1}{k} |\mathcal{S}(\phi_i) \cap \mathcal{S}(\psi_i)| \tag{11}$$

where $|\cdot|$ is the size of the intersection.

The choice to use mutual nearest-neighbors Our initial efforts to measure alignment with CKA revealed a very weak trend of alignment between models, even when comparing models within their own modality. This has also been observed by (Bansal et al., 2021), which had relied on alternative metrics such as model-stitching as it "reveals aspects of representations that measures such as centered kernel alignment (CKA) cannot" (Bansal et al., 2021).

We chose to use nearest-neighbor as a metric, as methods like CKA has a very strict definition of alignment, which may not fit our current needs. For instance, understanding the precise similarity between unrelated items, such as an orange and Bill Gates, may not be critical.

Relationship between CKA and Mutual Nearest-Neighbors Let $\phi_i \in \mathbb{R}^n$ and $\psi_i \in \mathbb{R}^m$ be vectorized features of two models (*e.g.* language and vision models). Let $\mathbf{K}_{ij} = \kappa(\phi_i, \phi_j)$ and $\mathbf{L}_{ij} = \kappa(\psi_i, \psi_j)$ be the kernel matrices computed from a dataset using some kernel-function κ. Using an inner-product kernel, the ij-th entry of the centered counterpart of these Kernel matrices is:

$$\bar{\mathbf{K}}_{ij} = \langle \phi_i, \phi_j \rangle - \mathbb{E}_l[\langle \phi_i, \phi_l \rangle] \qquad \bar{\mathbf{L}}_{ij} = \langle \psi_i, \psi_j \rangle - \mathbb{E}_l[\langle \psi_i, \psi_l \rangle] \tag{12}$$

Then, the cross-covariance of $\mathbf{K}$ and $\mathbf{L}$ is given by:

$$\mathsf{HSIC}(\mathbf{K}, \mathbf{L}) = \frac{1}{(n-1)^2} \mathsf{Trace}(\bar{\mathbf{K}}\bar{\mathbf{L}}) \tag{13}$$

which serves as an empirical estimator of the Hilbert-Schmidt Independence Criterion (Gretton et al., 2005). The Centered Kernel Alignment (CKA) (Kornblith et al., 2019) is then its normalized counterpart:

$$\mathsf{CKA}(\mathbf{K}, \mathbf{L}) = \frac{\mathsf{HSIC}(\mathbf{K}, \mathbf{L})}{\sqrt{\mathsf{HSIC}(\mathbf{K}, \mathbf{K})\mathsf{HSIC}(\mathbf{L}, \mathbf{L})}} \tag{14}$$

CKA measures the congruence between two random variables, with a maximum alignment of 1 and a minimum of 0. It is invariant to isotropic scaling and offers a strict notion of alignment, measuring alignment across all samples. Hence, the CKA score reflects the global similarities of the models. This can be illustrated by expanding the trace term in HSIC:

$$\mathsf{Trace}(\bar{\mathbf{K}}\bar{\mathbf{L}}) = \sum_i \sum_j \left(\langle \phi_i, \phi_j \rangle - \mathbb{E}_l[\langle \phi_i, \phi_l \rangle]\right)\left(\langle \psi_i, \psi_j \rangle - \mathbb{E}_l[\langle \psi_i, \psi_l \rangle]\right) \tag{15}$$

One can modify the definition of alignment to restrict the cross-covariance measurement to samples considered to be nearest neighbors of the current sample i. This emphasizes similarity over dissimilarity, biasing the measure toward local alignment:

$$\mathsf{Align}_{\mathsf{knn}}(\mathbf{K}, \mathbf{L}) = \sum_i \sum_j \alpha(i,j) \cdot \left(\langle \phi_i, \phi_j \rangle - \mathbb{E}_l[\langle \phi_i, \phi_l \rangle]\right)\left(\langle \psi_i, \psi_j \rangle - \mathbb{E}_l[\langle \psi_i, \psi_l \rangle]\right) \tag{16}$$

$$\text{where} \qquad \alpha(i,j) = \mathbb{1}[\phi_j \in \mathsf{knn}(\phi_i) \wedge \psi_j \in \mathsf{knn}(\psi_i) \wedge i \neq j] \tag{17}$$